Accurate on Average, Blind at the Fork

Anonymous Author(s)
Affiliation
Address
email

Abstract: A world model that predicts well on average can still be wrong exactly where it matters for safety: at forks, where small, realistic deviations send a manipulation task into different outcomes. We measure this *branching fidelity* with a goal-free fork probe modelled on the human intuitive physics engine [1]: sample the robot's measured execution and perception error, roll the intended action chunk forward 64 times, and flag the chunk if the futures split into distinct outcomes. With exact physics and a real camera-based perception model, the probe matches ground-truth fragility on unseen Jenga layouts (κ up to 0.84 against a ceiling of 0.92), also when outcomes are read from images. Replacing the simulator by a stochastic token-space world model that predicts futures $3\times$ better than a no-motion baseline drops agreement to κ 0.18: its sampled spread has the right size but ignores where futures branch. Robots that check actions before executing them need world models with branching fidelity, and average-error benchmarks do not measure it.

Keywords: World models, intuitive physics, runtime safety

1 Introduction

Modern manipulation policies, from diffusion policies to vision-language-action models, act by committing to short *action chunks* [2, 3, 4]. Before a chunk runs, a monitor can ask whether it is safe. In contact-rich tasks such as Jenga extraction the answer hinges on millimetres: the same command can slide the block out or tip a neighbour, depending on small errors in execution and perception. Such a chunk is precarious even if its nominal outcome is fine, because uncertainty the robot cannot control decides what happens.

People judge precariousness by running a few noisy mental simulations over an uncertain scene and checking whether the outcomes disagree, the *intuitive physics engine* (IPE) [1, 5, 6]. A robot can do the same, but it needs a model to simulate with. Outside simulation that model must be learned, and the debate on whether robots need world models then becomes concrete: a world model is useful for this check only if its rollouts *branch where the real dynamics branch*. Prediction error does not measure this.

Contributions. (1) A *fork probe* that checks an action chunk before execution without task labels, failure data or fitted thresholds, by sampling the robot's own measured execution and perception error (Sec. 2). (2) Validation of the probe with exact physics and real camera perception on unseen layouts, including reading outcomes from images, which sets the reference a world model should reach (Sec. 3.1). (3) *Branching fidelity* as a world-model evaluation: a token-space world model with low prediction error still loses the forks, and a diagnosis shows why (Sec. 3.2).

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2 Setup and the fork probe

Task. A simulated Franka Panda (MuJoCo) extracts blocks from Jenga-like stacks. At every chunk start a policy commits to an 8-step action chunk. Our chunks come from recorded extraction episodes, but the probe treats the policy as a black box and sees only the commanded chunk, one camera image and proprioception, so it applies to any chunked policy.

What counts as unsafe. A chunk is *fragile* if deviations the robot cannot control, its execution error and the error in its perception of the scene, can send the outcome into qualitatively different regimes. This needs no goal predicate or failure label, which a monitor around a generalist policy cannot assume; it isolates cases where intent no longer determines the outcome; and in contact-rich manipulation damage typically happens at such boundaries. A chunk that fails every time is not fragile and needs a task-level check.

The fork probe (Fig. 1a) imagines 64 versions of the chunk. Version j subtracts an execution-error snippet, drawn from the robot's measured tracking residuals (median chunk maximum 3.2 mm), and starts from a perturbed perceived scene. The scene is perceived by a frozen DINOv2 encoder [7] on one close-camera image with a keypoint head that regresses the poses of the three blocks in play (2.5 mm held-out error). The perception error model is measured once, offline: we run the perception model on held-out validation images with known poses and fit a zero-mean Gaussian to its errors (a full 9×9 covariance over block positions, which captures correlated errors, plus a rotation covariance per block). Each rollout starts from the estimate plus one draw from this model, as in the IPE, and ends with an imagined 30-step hold on the last command so the outcome settles. The probe flags the chunk if, at both times, the endings (object positions or image tokens) projected on their first principal axis prefer two Gaussians to one by BIC, are well separated (Ashman's $D > 2$ [8]), have at least 2 rollouts in the smaller group, and keep the same grouping. These are fixed statistical conventions, not fitted to labels. A variant hedges against an unlucky estimate by drawing 5 scene hypotheses and running the probe around each ("5 hypotheses", flag if at least 3 split).

Rollout engine. The probe needs a model to imagine the 64 futures and does not care which. Our reference engine is MuJoCo: the three blocks in play are placed at the perceived pose plus one error draw, while everything else (other blocks, geometry, masses, robot state) comes from the simulator, which is privileged information a real robot lacks. The learned engine is a world model that rolls out in image-token space from one camera image (Sec. 3.2); whether it can replace the simulator is the question of this paper.

Ground truth. One real outcome cannot grade a monitor: a fragile chunk may succeed by luck. Fragility is a property of the spread of possible outcomes, so we compute it with privileged access. *Fragility truth* runs the probe from the *true* scene in exact physics, with the same execution and perception error draws: the answer an oracle that knows the scene would give at the robot's own uncertainty. Two independent oracle runs agree at Cohen's κ 0.85–0.92, the ceiling any monitor can reach. Because this truth uses the same split test, we add a purely physical label: a *topple fork* is a state where some but not all oracle rollouts (2–62 of 64) knock over a neighbouring block. Each evaluation uses an unseen test set of stack layouts (up to 60 topple forks plus 100 safe states), under a protocol frozen before the data existed.

3 Results

3.1 Validating the probe with exact physics

On 112 development states with exact physics and the true scene, the split test flags all 23 topple forks and 10% of the 89 quiet states. With real perception, sampling the perception error is essential: perception errors of 2–4 mm are comparable to execution noise, and a probe that trusts the estimate ("no perception sampling") reaches only 38–55% of the ceiling κ, while the probe reaches 76–91%

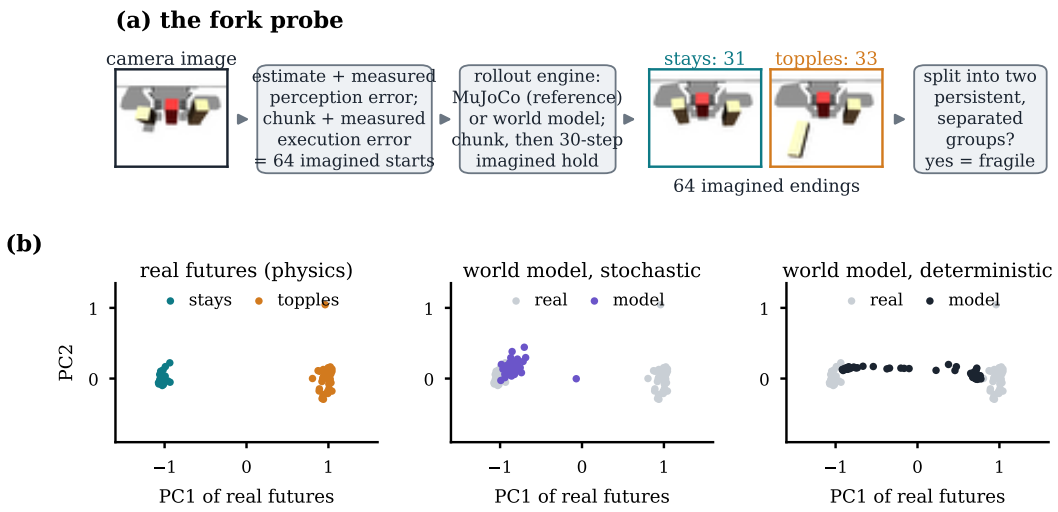


Figure 1: (a) The fork probe on one held-out state (33 of 64 real futures topple a neighbour). (b) The same state's futures in latent space, on the real futures' principal axes: physics splits into two groups; the stochastic world model stays near one outcome (most of its spread lies off this plane); the deterministic one spreads along the branch axis without a clean gap.

Table 1: Probe with MuJoCo and real perception on unseen test sets (κ vs fragility truth).

	Set A (153 states)		Set B (160)		Set C (156)	
outcomes read from object positions	κ	topple	κ	topple	κ	topple
ceiling (two oracle runs)	0.85	–	0.92	–	0.92	–
no perception sampling	0.32	–	0.51	0.65	0.42	0.67
fork probe	0.76	–	0.70	0.77	0.84	0.91
fork probe, 5 hypotheses	0.73	0.90	0.74	0.90	0.82	0.89
outcomes read from images: fork probe	–	–	0.70	0.77	0.77	0.88
5 hypotheses	–	–	–	–	**0.83**	**0.88**

(Table 1). Reading outcomes from DINOv2 tokens of rendered endings instead of object positions changes few decisions (κ 0.86–0.90 between the two), so the probe works from images alone: on test set C it agrees with fragility truth at κ 0.83 and catches 88% of topple forks (95% CI 0.77–0.94) with 7% false alarms on other states. This is the reference a world model should reach.

3.2 Branching fidelity of a learned world model

Outside simulation the privileged MuJoCo engine is unavailable, so a learned model must take its role, and the probe becomes its test: given one image and the 64 perturbed chunks, do its rollouts branch where physics branches? Since outcomes can be read from image tokens, a world model in token space can be probed exactly like the simulator. Our candidate encodes the image with DINOv2, compressed by a fixed per-patch PCA to a $28 \times 28 \times 8$ latent; the split test on this latent agrees with the position-based test at κ 0.93 on held-out real futures. A flow-matching transition predicts the next latent from the current one and the action, with fresh noise at every step, so repeated rollouts can differ. It is trained on 8 futures per image whose start scenes were sampled within the measured perception error, so the spread of its targets is the robot's own uncertainty. A deterministic version of the same network is the control. Both predict well on average, beating a no-motion baseline on 19-step latent error by 3–4×. Yet on test set C the stochastic model reaches κ 0.18 and catches 18% of topple forks, below its deterministic control (κ 0.42) and far below the simulator reference (κ 0.83; Table 2).

Table 2: World models on test set C, and diagnostics on 60 held-out states with 64 real futures each.

rollouts from	κ	topple	spread	main axis	fork ratio
reference (MuJoCo; real futures)	0.83	0.88	1.00	0.57	4.8
token world model, stochastic	0.18	0.18	0.98	0.15	1.3
token world model, deterministic	0.42	0.60	0.04	0.69	32
[TODO: iteration 2]					

Why the stochastic model fails. Table 2 compares, with real futures, the total spread of the 64 endings, the share of that spread on its main axis, and how much larger it is on fork states than on others. The stochastic model reproduces the *amount* of spread but not its *shape* or *placement*: its spread is isotropic and nearly the same on forks and non-forks, the average uncertainty of futures rather than where they branch; other samplers are worse. The deterministic model's spread is tiny but aligned with the branch axis and 32$\times$ larger on forks (Fig. 1b), and started from rendered perturbed scenes it doubles agreement with real futures (κ 0.17$\rightarrow$0.37). Per-step noise is not wrong in general (diffusion weather ensembles stay well dispersed [9]); here it was misplaced. As in the IPE, the uncertainty that matters is in the initial conditions, and roughly right dynamics amplify it along the branch. A second iteration (a learned operator perturbing the start latent by physical perception error, then deterministic dynamics) is under evaluation [TODO: result].

4 Related work

Simulation under uncertainty. Beyond the IPE and counterfactual accounts of physical support [10], robotics runs Monte Carlo rollouts over pose, friction or model parameters to score grasps and pushes [11, 12] or to build safety filters [13, 14]. Closest to our setup, Cannizzaro et al. [15] sample measured perception and placement noise through a simulator to predict whether a block tower stands, and Goyal [16] perturbs action chunks with policy-scale noise to measure, offline, how fast deviations grow. These score a task predicate, labels [17] or a growth rate; Zapolsky and Drumwright [18] proposed goal-free detection of divergent particle traces but evaluated task failure. We judge only whether the outcome set splits, a Monte Carlo test of proximity to a basin boundary [19].

Runtime monitors. Latent safety filters use learned failure classifiers, also on Jenga [20, 21]; failure detectors use policy uncertainty calibrated on successes [22, 23]. Both measure what a model does not know, not whether the physics is at a branch.

World models and their uncertainty. Learned simulators and latent world models [24, 25, 26, 27, 28] struggle with stiff contact [29], and audits find poor physical grounding and optimism under failure-inducing actions [30, 31, 32]. Yu et al. [33] propose evaluating world models on interventional action branches by per-branch error; our model has low error yet loses the forks. Stochastic and ensemble models [34, 35] are judged by likelihood or return, not by where their spread falls.

5 Conclusion and limitations

A robot can judge fragility the way people do, by sampling its own perception and execution uncertainty through a rough simulator. For this workshop's question, our results suggest that such checks need a queryable world model with branching fidelity, which our trained world model lacks despite low average error; here the uncertainty that matters enters through the initial conditions. Average-error benchmarks would rank this model well; branching fidelity is a needed additional axis.

Limitations. Simulation, one task, three tracked blocks; privileged MuJoCo geometry; one perception error model for all images; fragility truth shares the split test (topple forks are independent); test sets vary; single training runs; always-failing chunks are not flagged.

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